9.3.3 Removable and Jump Discontinuities

Let us look at the following functions :

(i) $f(x) = \dfrac{\sin x}{x}$

(ii) $g(x) = C(x)$, where $C(x)$ is as defined in Example 9.38.

The function $f(x)$ is defined at all points of the real line except $x = 0$. That is, $f(0)$ is undefined, but $\lim\limits_{x \to 0} \dfrac{\sin x}{x} = 1$ exists. If we redefine the function $f(x)$ as

$$h(x) = \begin{cases} \dfrac{\sin x}{x}, & x \neq 0 \\ 1, & x = 0 \end{cases}$$

h is defined at all points of the real line including $x = 0$. Moreover, h is continuous at $x = 0$ since

$$\lim_{x \to 0} h(x) = \lim_{x \to 0} \frac{\sin x}{x} = 1 = h(0).$$

Note that $h(x) = f(x)$ for all $x \neq 0$. Even though the original function $f(x)$ fails to be continuous at $x = 0$, the redefined function became continuous at 0. That is, we could remove the discontinuity by redefining the function. Such discontinuous points are called removable discontinuities. This example leads us to have the following.

Definition 9.10

A function f defined on an interval $I \subseteq$ is said to have removable discontinuity at $x_0 \in I$ if there is a function $h : I \to$ such that $h(x) = \begin{cases} f(x), & \text{if } x \neq x_0 \\ \lim\limits_{x \to x_0} f(x), & \text{if } x = x_0 \end{cases}$.

Note that for removable discontinuity, $\lim\limits_{x \to x_0} f(x)$ must exist.

Now if we examine the function $g(x) = C(x)$ (see Example 9.38), eventhough it is defined at all points of $[0, \infty)$, $\lim\limits_{x \to 100} g(x)$ does not exist and it has a jump of height $\lim\limits_{x \to 100^+} g(x) - \lim\limits_{x \to 100^-} g(x) = 16 - 14 = 2$, which is finite. Since $\lim\limits_{x \to 100} g(x)$ does not exist, it is not continuous at $x = 100$. Such discontinuities are called jump discontinuities. Thus we have the following :

Definition 9.11

Let f be a function defined on an interval $I \subseteq$. Then f is said to have jump discontinuity at a point $x_0 \in I$ if f is defined at x_0,

$\lim\limits_{x \to x_0^-} f(x)$ and $\lim\limits_{x \to x_0^+} f(x)$ exist but

$\lim\limits_{x \to x_0^-} f(x) \neq \lim\limits_{x \to x_0^+} f(x)$.

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